

Division of polynomials

Dividing by a monomial — term by term, and only when it goes exactly.

LESSON 75–76

2 × 40 MIN

MATEMATIKA 7, §35

S8 2.3

LESSON CLOCK

15'

25'

25'

20'

5'

Term by term	15 min
When it works	25 min
Reversing a multiplication	25 min
Simplifying fractions	20 min
Homework	5 min

LEARNING OBJECTIVES

- Divide a polynomial by a monomial, term by term.
- Decide when the quotient is again a polynomial.
- Use division to reverse a multiplication.
- Simplify an algebraic fraction whose denominator is a monomial.

TEXTBOOK

National	pp. 212–217
Cambridge	Stage 8, pp. 18–24
Workbook	Exercise 2.3

01

Explanation

Term by term

$$\frac{a + b + c}{k} = \frac{a}{k} + \frac{b}{k} + \frac{c}{k}$$

Divide each term of the polynomial by the monomial.

QUOTIENT	TERM BY TERM	ANSWER
$(6x^2 + 9x) \div 3x$	$\frac{6x^2}{3x} + \frac{9x}{3x}$	$2x + 3$
$(8a^3 - 4a^2) \div 4a^2$	$2a - 1$	$2a - 1$
$(10x^3y - 15x^2y^2) \div 5x^2y$	$2x - 3y$	$2x - 3y$

EVERY TERM IS DIVIDED, INCLUDING THE LAST

$(8a^3 - 4a^2) \div 4a^2$ is $2a - 1$, not $2a$. The second term gives -1 , and dropping it is the standard slip.

When it works

The quotient is a polynomial only if the monomial divides **every** term exactly — the coefficients and each letter.

DIVISION	POLYNOMIAL?	WHY
$(6x^2 + 9x) \div 3x$	yes	$3x$ divides both terms
$(6x^2 + 9x) \div 3x^2$	no	$3x^2$ does not divide $9x$
$(4a + 6b) \div 2$	yes	2 divides both terms
$(4a + 5b) \div 2$	no	2 does not divide $5b$

THE DIVISOR MUST BE A COMMON FACTOR

Finding the highest common factor of all the terms tells you at once what the polynomial can be divided by — and that is exactly what factorising will use two lessons from now.

Reversing a multiplication

Division undoes multiplication, so every division can be checked by multiplying back.

DIVISION	ANSWER	CHECK
$(6x^2 + 9x) \div 3x$	$2x + 3$	$3x(2x + 3) = 6x^2 + 9x$ ✓
$(10a^2 - 5a) \div 5a$	$2a - 1$	$5a(2a - 1) = 10a^2 - 5a$ ✓

ONE MULTIPLICATION CHECKS THE WHOLE DIVISION

It takes ten seconds and catches every slip, including the dropped last term. There is no faster check in algebra.

Simplifying fractions

An algebraic fraction is simplified by cancelling the common factors of numerator and denominator.

FRACTION	SIMPLIFIED
$\frac{12a^3}{4a}$	$3a^2$
$\frac{6x^2 + 9x}{3x}$	$2x + 3$
$\frac{5ab}{10a^2}$	$\frac{b}{2a}$

CANCEL FACTORS, NEVER TERMS

In $\frac{6x^2 + 9x}{3x}$ the whole numerator is divided. Cancelling the x of $9x$ against the denominator alone, and leaving the $6x^2$ untouched, is wrong — the division applies to every term or to none.

02

Worked examples

EXAMPLE 1 Compute $(6x^2 + 9x) \div 3x$.

1

$\frac{6x^2}{3x} = 2x$

2

$\frac{9x}{3x} = 3$

3

$= 2x + 3$

4

Check: $3x(2x + 3) = 6x^2 + 9x$ ✓

ANSWER

$2x + 3$

EXAMPLE 2 Compute $(10x^3y - 15x^2y^2) \div 5x^2y$.

① First term: $\frac{10x^3y}{5x^2y} = 2x$.

② Second: $\frac{-15x^2y^2}{5x^2y} = -3y$.

③ $= 2x - 3y$

④ Check: $5x^2y(2x - 3y) = 10x^3y - 15x^2y^2 \checkmark$

ANSWER

$$2x - 3y$$

EXAMPLE 3 Is $(6x^2 + 9x) \div 3x^2$ a polynomial?

① First term: $\frac{6x^2}{3x^2} = 2 \checkmark$

② Second: $\frac{9x}{3x^2} = \frac{3}{x}$.

③ That is not a monomial.

④ No — the quotient is not a polynomial.

ANSWER

No

03 Interactive model

Ask the class to check every division by multiplying back, and mark only the checks for one lesson; the accuracy of the divisions rises immediately.

Cancelling common factors

$$\frac{6x^2 + 9x}{3x}$$

① $\frac{6x^2}{3x} = 2x$

The
first
term.

② $\frac{9x}{3x} = 3$

The
second
— it
does
not
vanish.

③ $2x + 3$

Check:

$$\begin{aligned} &3x(2x \\ &+ 3) \\ &= \\ &6x^2 \\ &+ 9x \\ &\checkmark \end{aligned}$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Division	Bo'lish	Деление
Quotient	Bo'linma	Частное
Term by term	Hadma-had	Почленно
Exactly divisible	Butun bo'linadi	Делится нацело
Common factor	Umumiy ko'paytuvchi	Общий множитель
To reverse	Teskari amal	Обратное действие
Algebraic fraction	Algebraik kasr	Алгебраическая дробь
Check	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

To divide a polynomial by a monomial:

divide the first term

divide every term

multiply

cancel one letter

$(8a^3 - 4a^2) \div 4a^2$ equals:

$2a$

$2a - 1$

$2a^3$

$2a - 4$

$(6x^2 + 9x) \div 3x$ equals:

$2x$

$2x + 3$

$2x + 9$

$2x^2 + 3$

$(6x^2 + 9x) \div 3x^2$ is:

a polynomial

not a polynomial

a monomial

zero

A division is checked by:

dividing again

multiplying back

estimating

nothing

In an algebraic fraction you may cancel:

terms

factors

anything

signs

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $(6x^2 + 9x) \div 3x$

ANSWER $2x + 3$

2. $(8a^3 - 4a^2) \div 4a^2$

ANSWER $2a - 1$

3. $(4a + 6b) \div 2$

ANSWER $2a + 3b$

4.
$$\frac{12a^3}{4a}$$

ANSWER $3a^2$

5. $(10a^2 - 5a) \div 5a$

ANSWER $2a - 1$

6. $(x^2 + x) \div x$

ANSWER $x + 1$

7. $(9y^3 + 3y) \div 3y$

ANSWER $3y^2 + 1$ **Medium · the standard question**

7 PROBLEMS

1. $(10x^3y - 15x^2y^2) \div 5x^2y$

ANSWER $2x - 3y$

2. $(12a^2b + 8ab^2) \div 4ab$

ANSWER $3a + 2b$

3. $\frac{5ab}{10a^2}$

ANSWER $\frac{b}{2a}$

4. Is $(6x^2 + 9x) \div 3x^2$ a polynomial?

ANSWER No

5. Is $(4a + 5b) \div 2$ a polynomial?

ANSWER No

6. Check $2x + 3$ by multiplying by $3x$

ANSWER $6x^2 + 9x$

7. $(a^3 - a^2) \div a^2$

ANSWER $a - 1$
Hard · stretch and reason

7 PROBLEMS

1. $(15x^4y^2 - 10x^3y^3 + 5x^2y) \div 5x^2y$

ANSWER $3x^2y - 2xy^2 + 1$

2.
$$\frac{4a^2b - 6ab^2}{2ab}$$

ANSWER $2a - 3b$

3. What must $3x$ be multiplied by to give $12x^3 - 9x^2$?

ANSWER $4x^2 - 3x$

4.
$$\frac{x^2 - x}{x}$$

ANSWER $x - 1$

5. The HCF of $6x^2$ and $9x$

ANSWER $3x$

6. Divide $8a^3b - 12a^2b^2 + 4ab$ by its HCF

ANSWER $2a^2 - 3ab + 1$

7. Why must the divisor be a common factor?

ANSWER Otherwise some term gives a fraction

07 Homework

Homework — 5 tasks

Check every division by multiplying the answer by the divisor.

1. Compute $(12x^2 - 8x) \div 4x$.

2. Compute $(9a^3b - 6a^2b^2) \div 3a^2b$.

3. Simplify $\frac{20m^3}{5m}$.

4. Is $(10x^2 + 4x) \div 4x^2$ a polynomial? Explain.

5. Find the HCF of $8a^2b$ and $12ab^2$, and divide their sum by it.